

Nicolas Sarmiento

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Education

Auburn University, Auburn, AL

M.S. Artificial Intelligence Engineering

Expected Dec. 2027

Incoming Graduate Student

Auburn University, Auburn, AL

B.S. Software Engineering & B.S. Aerospace Engineering

Expected Dec. 2026

GPA: 3.65/4.00

Certificate: Artificial Intelligence Engineering | **Coursework:** AI, Machine Learning, Federated Learning, Data Structures & Algorithms, Operating Systems

Technical Skills

Languages: Python, C#, SQL, MATLAB | **AI/ML:** PyTorch, TensorFlow, scikit-learn, Hugging Face, Flower

Backend/Data: FastAPI, Flask, ASP.NET Core, SQLAlchemy, REST APIs, ChromaDB | **Tools:** Git, Docker, GitHub Actions, pytest, ONNX Runtime

Projects

TensorFlow-RAG | Python, FastAPI, ChromaDB, ONNX, Claude API, Docker

GitHub

- Developed a **Retrieval-Augmented Generation (RAG)** web application over official TensorFlow documentation, integrating FastAPI, ChromaDB, and the Claude API.
- Reduced idle memory usage by **61% (465 MB to 180 MB)** by replacing the PyTorch embedding stack with ONNX Runtime, and built a **23-question** CI testing suite with pytest and GitHub Actions.

FedFighter | Python, PyTorch, Flower, CNN-LSTM, Federated Learning

GitHub

- Built a **federated intrusion-detection system** on the CICIDS2017 dataset using PyTorch and Flower, implementing a hybrid 1D-CNN/LSTM architecture. using PyTorch and Flower with a hybrid **1D-CNN/LSTM** architecture
- Evaluated FedAvg vs. FedProx across **5-client, 10-round non-IID experiments**, achieving a **92.4% classification accuracy** with FedAvg.

Weather Dashboard Web App | C#, ASP.NET Core, REST APIs, HttpClient

GitHub

- Created an **ASP.NET Core web application** to display real-time weather data using the OpenWeather REST API.
- Handled asynchronous data fetching with HttpClient, implemented geographic lookups, and secured API keys using environment variables.

Financial Sentiment Analyzer | Python, FastAPI, FinBERT, Hugging Face, Plotly

GitHub

- Designed a FastAPI service that uses **FinBERT** to classify financial news headlines by sentiment (positive, negative, neutral) with confidence scoring.
- Integrated the **TwelveData REST API** to plot sentiment predictions against **30-day stock performance** trends using Plotly.

Experience

Undergraduate Teaching Assistant, Auburn University – Auburn, AL

Aug. 2025 – Present

- Support **3 lab sections totaling 110+ students** in Python and MATLAB, teaching programming fundamentals, debugging, and computational problem solving.
- Diagnose code errors and provide individualized technical guidance while collaborating with instructors to reinforce course concepts.

Academic Support Peer Tutor, Auburn University – Auburn, AL

Sep. 2024 – Aug. 2025

- Delivered **125+ hours** of one-on-one instruction to **44 students** in Calculus I & II, Linear Algebra, and Differential Equations.
- Earned a **4.96/5.00 average rating across 51 reviews**, demonstrating consistent communication and instructional effectiveness.